Group X Progress Report:
My Group's Project Name

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1 Introduction

Pneumonia is a severe respiratory infection causing significant morbidity and mortality worldwide, especially in young children. Accurate and timely diagnosis is crucial for effective treatment and improving outcomes. Traditionally, diagnosis relies on expert radiologists interpreting chest X-rays, but deep learning-based automated detection systems offer potential for faster and scalable assessment. Our project aims to develop a binary classification model that distinguishes between normal and bacterial pneumonia pediatric chest X-ray images using convolutional neural networks (CNNs) and transfer learning with pretrained models.

2 Related Work

We use the publicly available pediatric chest X-ray dataset sourced from Guangzhou Women and Children's Medical Center (**?**). This dataset contains X-ray images labeled as Normal, Viral Pneumonia, and Bacterial Pneumonia. For our project, we focus on distinguishing Normal images from those showing Bacterial Pneumonia, making it a binary classification task.

The dataset is split into training, validation, and test sets in proportions of 80%, 10%, and 10% respectively. Stratified sampling was applied to ensure that each subset maintains the same class distribution as the overall dataset. Preprocessing involves resizing all images to a uniform dimension of 224×224 pixels to match the input size requirements of our convolutional neural network. Pixel intensities are normalized according to ImageNet's channel mean and standard deviation statistics to stabilize model training. To increase the diversity of training samples and reduce overfitting, we apply several data augmentation techniques. These include random resized cropping to simulate varying image scales, random horizontal flips to represent left-right invariance in X-rays, slight random rotations to account for variance in patient positioning, and brightness jittering to mimic variations in imaging conditions. Additionally, we introduced an optional histogram equalization step prior to other augmentations. This step enhances image contrast, making critical features more prominent to the network during training.

This preprocessing strategy collectively improves the robustness and generalization capacity of the trained model. As shown in Figure 1, our training set has a noticeable class imbalance, with the pneumonia class being significantly more common than the normal class. This imbalance was addressed by applying class weighting in the loss function.

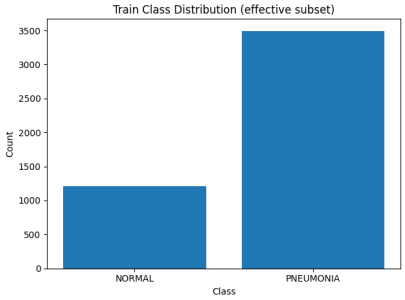


Figure 1: Train class distribution for the effective data subset. The dataset is imbalanced, with the pneumonia class containing nearly three times as many examples as the normal class. This motivates our use of class weights during model training.

3 Features

Our model processes the input chest X-ray images by first applying augmentations and normalization to prepare the data for training. The images are originally grayscale but are converted into three-channel format to be compatible with the pretrained ResNet18 model we use. The network learns features automatically through fine-tuning the pretrained model. This means the model ad-

justs both low-level and higher-level layers to better capture patterns specific to pneumonia detection. We do not perform any manual feature engineering beyond the applied data augmentations. The representations are learned end-to-end, enabling the model to identify relevant structures and textures within the X-ray images that distinguish normal lungs from bacterial pneumonia.

4 Implementation

The model uses a ResNet18 convolutional neural network pretrained on ImageNet. We start by freezing all the layers of the pretrained network except the last fully connected layer, which we replace to adapt the model for the binary pneumonia classification task. In this initial phase, we train only the new classification layer, allowing the network to learn how to map extracted features to our target labels while keeping the underlying feature extraction stable. Once the classifier head reaches a satisfactory performance, we unfreeze the last convolutional block to fine-tune deeper layers. This allows the network to adjust more specialized features relevant to pediatric chest X-rays and pneumonia detection. The fine-tuning uses a lower learning rate to ensure stable training without overwriting useful pretrained features. We employ weighted cross-entropy loss during training to mitigate class imbalance by giving more importance to the minority pneumonia class. The Adam optimizer is used with weight decay regularization to help prevent overfitting by penalizing large parameter values. Additionally, we employ early stopping based on validation F1-score, which halts training when no improvement is observed for a set number of epochs, saving the best model weights. For baseline comparison, we implement a simple majority-class classifier that always predicts the most frequent class in the training set. This baseline provides a minimal performance benchmark that our model should outperform. During training, we incorporate data augmentation and preprocessing to enhance model robustness. We encountered typical challenges such as tuning the balance between frozen and unfrozen layers and setting appropriate learning rates. These were addressed through iterative experimentation, monitoring training and validation metrics to guide hyperparameter selection. The implementation is modular to allow for easy experimentation with other models, loss functions, or augmentation strategies in the future. So far, the approach demonstrates an effective transfer learning strategy well-suited for medical image classification tasks like pneumonia detection.

5 Results and Evaluation

We evaluate our model using stratified training, validation, and test splits. The validation set guides hyperparameter tuning and early stopping, while the test set provides an unbiased assessment of final model performance.

Our evaluation metrics include accuracy, precision, recall (sensitivity), F1-score, and ROC-AUC. Accuracy measures the overall correct predictions; precision and recall focus on positive (pneumonia) detection performance, which is critical in a medical context to minimize missed diagnoses. The F1-score balances precision and recall, and the ROC-AUC quantifies the model's discrimination ability across thresholds.

The baseline model used for comparison is a majority-class classifier that predicts the most frequent class in the training data.

From our results, the CNN model substantially outperforms the baseline across all metrics. Specifically, the validation F1-score improved from approximately 0.55 (baseline) to over 0.85 with the CNN. Recall increased notably, signaling fewer missed pneumonia cases. Confusion matrices further confirm enhanced true positive detection while controlling false positives. These preliminary results demonstrate that fine-tuning the pretrained ResNet18 with our preprocessing and augmentation pipeline yields a robust pneumonia classifier with strong generalization to unseen data. Future milestones include implementing k-fold cross-validation for more reliable estimates and further interpreting misclassification patterns to refine the model.

Write about your plans for the remainder of the project. This should include a discussion of the feedback you received from your TA, and how you plan to improve your approach. Reflect on your implementation and areas for improvement. Refer to item 6. This may be around 0.5 pages.

6 TA Feedback and Reflection

The primary feedback from our TA emphasized the importance of using a simple baseline for comparison, specifically the majority-class classifier. This baseline predicts the most frequent class in the training data and establishes a clear performance

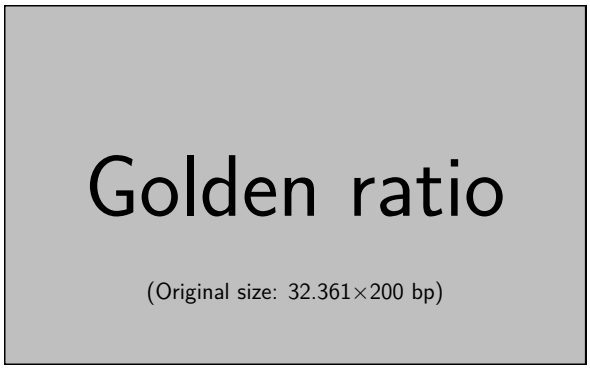


Figure 2: A figure with a caption that runs for more than one line. Example image is usually available through the `mwe` package without even mentioning it in the preamble.

threshold that the CNN model should surpass. Including such a baseline is essential for contextualizing model results and avoiding overstated performance claims.

Reflecting on this advice, we integrated the majority baseline into our evaluation framework and confirmed that our CNN model consistently outperforms it across key metrics. This strengthens confidence in the effectiveness of our transfer learning approach for pneumonia detection.

The TA also recommended focusing on binary classification between Normal and Bacterial Pneumonia instead of including Viral Pneumonia. This simplifies the task and keeps the project scope focused. All data preparation and model design follow this two-class goal, with the majority baseline serving as a clear reference point. A multiclass version may be explored later as an extension to this project.

Overall, the TA's guidance reinforced the importance of rigorous benchmarking and comparative evaluation, which will remain central to our approach in future milestones.

7 Template Notes

7.1 Tables and figures

See Table 1 for an example of a table and its caption. See Figure 2 for an example of a figure and its caption.

7.2 Citations

Table 1 shows the syntax supported by the style files. We encourage you to use the natbib styles. You can use the command `\citet` (cite in text) to get "author (year)" citations, like this citation to a paper by Gusfield (1997). You can use the command `\citep` (cite in parentheses) to get "(author, year)" citations (Gusfield, 1997). You can use the command `\citealp` (alternative cite without parentheses) to get "author, year" citations, which is useful for using citations within parentheses (e.g. Gusfield, 1997).

7.3 References

Many websites where you can find academic papers also allow you to export a bib file for citation or bib formatted entry. Copy this into the `custom.bib` and you will be able to cite the paper in the LaTeX. You can remove the example entries.

7.4 Equations

An example equation is shown below:

$$A = \pi r^2 \tag{1}$$

Labels for equation numbers, sections, subsections, figures and tables are all defined with the `\label{label}` command and cross references to them are made with the `\ref{label}` command. This an example cross-reference to Equation 1. You can also write equations inline, like this: $A = \pi r^2$.

Team Contributions

Write in this section a few sentences describing the contributions of each team member. What did each member work on? Refer to item 7.

References

Rie Kubota Ando and Tong Zhang. 2005. A framework for learning predictive structures from multiple tasks and unlabeled data. *Journal of Machine Learning Research*, 6:1817–1853.

Galen Andrew and Jianfeng Gao. 2007. Scalable training of L1-regularized log-linear models. In *Proceedings of the 24th International Conference on Machine Learning*, pages 33–40.

Dan Gusfield. 1997. *Algorithms on Strings, Trees and Sequences*. Cambridge University Press, Cambridge, UK.

Mohammad Sadegh Rasooli and Joel R. Tetreault. 2015. Yara parser: A fast and accurate dependency parser. *Computing Research Repository*, arXiv:1503.06733. Version 2.

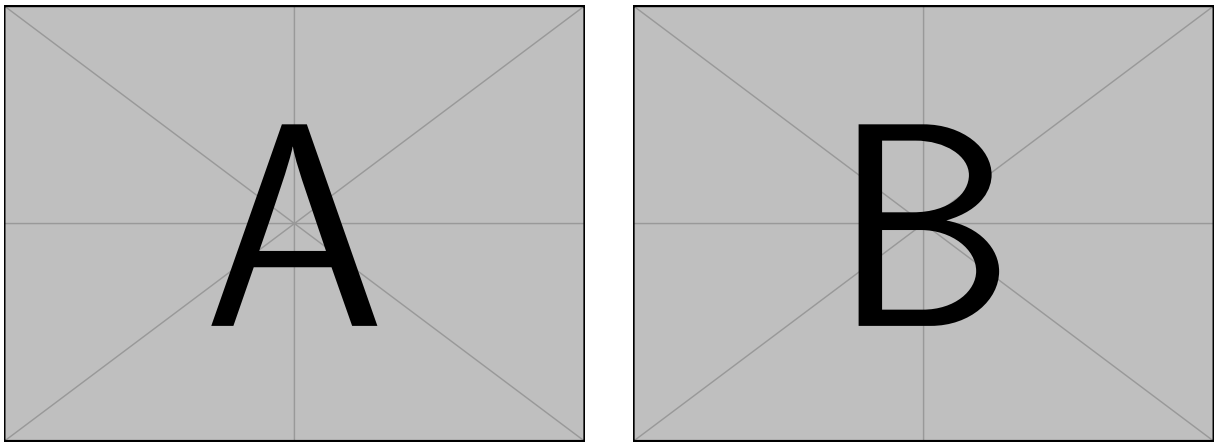


Figure 3: A minimal working example to demonstrate how to place two images side-by-side.

Output	natbib command	ACL only command
(Gusfield, 1997)	`\citep`	
Gusfield, 1997	`\citealp`	
Gusfield (1997)	`\citet`	
(1997)	`\citeyearpar`	
Gusfield's (1997)		`\citeposs`

Table 1: Citation commands supported by the style file.